Portable Generator

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Portable Generator (Optional)

What it is used for

A portable generator produces electrical power from an engine-driven alternator for temporary use. Correct setup and operation help prevent electric shock, fire, carbon monoxide poisoning and equipment damage.

Main parts

- Identify the engine and alternator housing.
- Locate the fuel tank, fuel cap and fuel shut-off control.
- Identify the engine start and stop controls.
- Locate the AC outlets, circuit breakers and grounding terminal, where fitted.
- Identify the voltage and frequency indicators, if provided.
- Locate the engine oil filler, dipstick and air filter.
- Identify the exhaust outlet, frame and transport handles.

Before you start

1. Read the operating instructions and confirm the generator's rated output.
2. Operate the generator outdoors, far from doors, windows and air intakes.
3. Never run the generator inside homes, classrooms, garages or enclosed spaces.
4. Inspect fuel lines, electrical cables, outlets and housing for damage or leaks.
5. Check engine oil and fuel levels according to manufacturer instructions.
6. Place the generator on a firm, dry and level surface away from flammable materials.
7. Ensure connected loads do not exceed the generator's rated capacity.
8. Disconnect electrical loads before starting the generator.

How to use

1. Position the generator outdoors in a well-ventilated area with exhaust directed away from people.
2. Check that the generator is switched off and all connected loads are disconnected.
3. Refuel only when the engine is cool and switched off, following the instructions.
4. Open the fuel valve and set the choke or starting control as specified.
5. Start the engine using the designated starting method.
6. Allow the engine to stabilise and check the voltage or status indicators.
7. Connect suitable loads using correctly rated cables and protected outlets.
8. Switch on connected equipment in stages while monitoring the total load.
9. Watch for overheating, unusual noise, fuel leakage or unstable output.

10. Disconnect loads before switching off the generator and closing the fuel valve.

⚠ Safety

- Never operate a generator indoors or in partially enclosed spaces because carbon monoxide can be fatal.
- Keep the generator dry and never connect loads with wet hands.
- Never refuel a running or hot engine or smoke near fuel.
- Use correctly rated cables and suitable residual-current protection where required.
- Do not connect the generator directly to building wiring without an approved transfer arrangement installed by a qualified electrician.
- Stop the generator immediately if fuel leaks, sparks or dangerous electrical faults occur.

Care & storage

- Switch off the engine and allow it to cool before cleaning or servicing.
- Check engine oil, air filter and spark plug according to the maintenance schedule.
- Inspect outlets, cables, fuel lines and frame regularly for damage.
- Store the generator dry with fuel handled according to manufacturer guidance.

Common mistakes to avoid

- Running the generator indoors or near open windows and ventilation intakes.
- Overloading the generator with appliances beyond its rated output.
- Refuelling while the engine is hot or running.
- Connecting the generator to building wiring without approved isolation equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.